

## PROJECT PLAN

### Diet Suggestion System for children under the age of 5 in Kilifi County

#### Requirements

1. User requirements
2. Data collection (Surveys, interviews)
3. Health data collection
4. Determination, dedication and time

#### Who are the intended users of this system

1. Caregivers
2. Parents
3. Hospitals
4. Guardians
5. Hotels

#### How they are expected to use it

Any user of the system will need to create an account in which he/she will add the details of the child or the children he/she intends to manage their diet.

Due to critical information like health data, there will be several layers of security e.g., setting strong passwords and 2FA authentication through email.

Any user will need to log in to access the system services at all times.

#### What the users will be able to modify, edit, delete or add to the system

1. The users upon successful logging will be able to update his/her child details e.g health conditions, Allergies, Favorite meals and age.
2. The user will be able to add meals which they feel they need and may not be in the system. These meals will only help in scheduling of meals for the specific user and will not affect the entire system verified meals.
3. The users will be able to modify their passwords.
4. Users will be able to clear their data when they deem fit

#### Expected parts of the system

1. Being a web app, the system will have **unauthenticated** landing page, this page will be called **home page**. This page will be accessible to everyone on the internet in Kilifi county. The page will consist of a beautiful UI about Diet recommendation, a contact section, a button linked to a login page, and one linked to a registration page
2. **Registration page**. This will be the first step to use the system. Every user will need to register via this page. This page is intended to have a beautiful and attractive UI.
3. **A login page**. This page will be used for authentication of users during login. It will also have a beautiful UI
4. **Password reset page**. This page will be used for helping with forgot password issues
5. **Dashboard**. This will be **authenticated** landing page immediately after logging in.

### Expected features (functionalities) and parts of dashboard

- i. A modern and responsive navbar with beautiful nav links and buttons for navigating the system
- ii. A profile section with the necessary details of the user and some buttons to allow some features like data updates, subscriptions and data deletion.
- iii. **There will be different sections in this page. The sections will include:**
  - a. **A meal recommendation section.** This is the section where a user will be able to select the meal to recommend for the day e.g Breakfast, Lunch, Supper. This recommendation will depend on the child's provided data and past recommendations if any.
  - b. **Analysis section.** This section will contain analysis about how the user has been using the system for diet recommendations. It will include a dynamic bar chart showing the meals and their names taken by a user on a daily basis, weekly basis, monthly basis and yearly basis for a period of up to 5 years. This section will also show in percentages, nutrition progress e.g. Vitamin taken in percentage compared to target
  - c. **Meal scheduling section.** This section will be used to schedule and access scheduled meals. The users will be able to schedule a meal for the coming days to help them budget and so on. They will also be able to cancel any scheduled when they wish.
  - d. **Notifications section.** This section will contain in-app notifications like reminders when a scheduled meal date has reached, any alerts and other reminders about meals.
6. **Details update page (Authenticated).** This page will be used to confirm and update child details in the database for better recommendation.
7. **Premium subscription page (Authenticated).** This page will be used for premium subscription. It will include selection, confirmation and payment of the premium selected.
8. **Meal update page (Authenticated).** This page will allow only admins (Verified health practitioners) to add and delete meals from the system

### Preparation (Basic Gantt Chart)

| Day                              | Activity                                                                                                                  |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Day 1- day 2                     | Development of a landing page (Home page)                                                                                 |
| Day 3 – day 5                    | Development of login, password reset page and registration pages                                                          |
| Day 6                            | Creation of a database (MySQL database). Designing the schemas and tables                                                 |
| Day 7 – day 8                    | Creation of login, password reset and registration routes and server file. Also installation of all node modules required |
| Day 9                            | Creating a connection pool and linking the registration, password reset and login pages to the database                   |
| Day 10                           | Working on the Dashboard UI                                                                                               |
| Day 11                           | Working on profile section of the dashboard                                                                               |
| Day 12 – day 14                  | Working on a meal recommendation section of the dashboard                                                                 |
| Day 15 -day 16                   | Working on the analysis section of the dashboard                                                                          |
| Day 17                           | Working on the meal scheduling section of the dashboard                                                                   |
| Day 18 -day 19                   | Working on the notification section of the dashboard                                                                      |
| Day 20                           | Working on details update page                                                                                            |
| Day 21                           | Working on of the premium subscription Page                                                                               |
| Day 22 – day 23                  | Working on the meal update page                                                                                           |
| Day 24 – Day 27                  | Testing, making corrections and improvements                                                                              |
| Total least time needed: 1 Month |                                                                                                                           |